

Dhruv Madan

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Education

Worcester Polytechnic Institute (WPI)

B.S. Robotics Engineering & B.S. Computer Science

Worcester, MA | May 2027

GPA 4.0 - 4x Dean's List Awardee

Current Relevant Coursework: Sensing & Perception in Robotics, Advanced Actuation in Robotics, Technical Writing, Robotic Navigation & Sensing, Advanced Robotic Motion Planning, AI for Robotics, Vision-Based Robotic Manipulation, Industrial Robotics

Skills & Interests

Programming: C/C++, Matlab/Simulink, Python, Git, Java, ROS/ROS2, Linux, Gazebo, OMPL, Genesis, Docker, PLC

Software: SolidWorks, Autodesk Inventor, OpenSCAD, Microsoft Office, KiCAD, Grafana, Minitab, JIRA, Robot Studio

Fabrication & Machining: 3D printing, Laser cutting, CNC, CAM, FANUC Robotics, Universal Robotics, Oscilloscope, DMM

Interests: Photography, Hiking, Biking, Running, Baseball, F1, Industrial Robotics, Mobile Robots

Professional Experience

Software Engineering Intern, Brooks Automation

Chelmsford, MA | May 2025 - Present

- Architected a PostgreSQL database to centralize experiment data, reducing data retrieval time by **~40%** and eliminating **100+ fragmented local log files** used previously.
- Built data-pipeline tools for uploading, parsing, and auto-generating performance reports, cutting manual analysis time by **60–70%** and enabling **same-day insights** on robot experiments.
- Deployed backend services to **AWS (EC2 & RDS)**, improving system uptime, scalability, & accessibility for engineers.
- Engineered an integrated hardware–software interface for safety light curtains, ensured seamless robot integration and improved operator safety by ensuring **100% fail-safe detection** during robot operation.

Teaching Assistant; Embedded Systems for Robotics, WPI

Worcester, MA | Jan 2025 - Present

- Guided **50+** students in programming microcontrollers, developing electronic circuits, and debugging hardware/software integration while fostering a collaborative learning environment.
- Developed technical documentation and troubleshooting guides for lab equipment used by **300+** students, enhancing accessibility and streamlining lab workflows.

Robot Technician, Starship Technologies

Worcester, MA | Feb 2024 - Present

- Maintained fleet operational efficiency through rigorous troubleshooting and repairs. Increased fleet uptime by **50%**.
- Enhanced fleet performance by monitoring systems, conducting rigorous testing, and performing preventive maintenance, maintaining fleet health **above 90%**.

Projects

Vision-Based Robotic Grasping Quality Pipeline

Aug 2025 - Oct 2025

- Engineered a vision-based robotic grasp-quality estimation system using OpenCV, increasing grasp stability by **~30%** compared to centroid-based baselines in course evaluations.
- Developed a modular computer-vision pipeline that extracts object contours, computes geometric features, and ranks grasp candidates, achieving **>90% correct grasp-point identification** on common test cases.
- Integrated pipeline into a modular architecture suitable for ROS-based manipulation tasks on arbitrary objects.

Autonomous Navigation & Exploration Robot

Mar 2025 – May 2025

- Implemented SLAM and localization using LiDAR data in ROS, enabling autonomous mapping of an environment with **>95%** positional accuracy.
- Developed an A* path planner to compute collision-free routes, reducing navigation time to waypoints by **~30%** compared to baseline methods.
- Programmed a pure pursuit controller that improved trajectory tracking accuracy by **20%** and ensured smooth robot motion.
- Integrated ROS nodes for perception, planning, & actuation into a complete navigation stack, achieving fully autonomous operation.

4-DOF Robotic Arm Manipulator with Object Detection

Jan 2025 - Mar 2025

- Designed, programmed, and validated a fully autonomous **4-DOF** robotic arm capable of sorting objects with high precision.
- Developed Forward/Inverse Kinematics, Velocity Kinematics & Trajectory Generation to enable smooth & accurate motion.
- Implemented real-time vision-based control systems, achieving **95%** detection and tracking accuracy via a camera-guided feedback loop.

Leadership & Activities

Leadership: Treasurer, IEEE WPI Student Branch - Public Relations Chair, PBE (Robotics Honor Society)

Eagle Scout, Boy Scouts of America